

GUIDELINES FOR EPN STATIONS

EPN Central Bureau
C. Bruyninx

Version Sept. 1, 2025

Available from the EPN Central Bureau at

https://epncb.oma.be/documentation/guidelines/guidelines_station_operationalcentre.pdf

Updates:

Sept. 1, 2025

- Removal of Operational Centres because obsolete
- Clarification of station labels in use
- Update on RINEX 4 headers
- Update of real-time section (e.g., exclude RTCM 2, mountpoint naming convention, minor textual changes)

March 22, 2022

- Updated links
- Submission of RINEX 3 is mandatory
- New section on submission of high-rate data

Oct. 4, 2021

- Updated link to NTRIP software

Oct. 4, 2019

- RINEX 2 submission optional if RINEX 3 is submitted
- ETRS89 coordinates in real-time streams
- Creation of RINEX files from receiver raw data
- Update of several links
- Submission of RTCM 3 preferred above RTCM 2
- Inclusion of description of MSM message types
- Updated link to ECGN Guidelines
- Remove submission of .SUM file to data centres

April 11, 2017: Regional data centre OLG is replaced by new data centre BEV in data flow figures.

Oct. 26, 2015

- Section 2.4.1: updated method for measuring connection between station marker and levelling network
- Section 3.2: Stations tracking more than two frequencies or satellite systems in addition to GPS and GLONASS must submit data files in both the RINEX 2 (dual frequency GPS and GLONASS) and RINEX 3 format (GPS, GLONASS plus other GNSS, SBAS, etc.). Removal of the '.flg' convention for data transmission.

May 28, 2013: Major revision and restructuring reflecting current state-of-the art.

June 13, 2012: Modified links due to the EPN CB web site restructuring

Feb. 1, 2012: Correction of description of stream

March 8, 2010:

- Section 2.1: updated 2.1.2, 2.1.14 & 2.2.0, added 2.1.19, 2.1.20, 2.1.22 & 2.1.25, updated 2.1.27, added 2.1.29, related to absolute antenna calibrations and multi-GNSS antenna
- Section 2.2: updated 2.2.2

- Section 2.3: updated 2.3.1 for on-line station site log submission tool
- Section 2.4: updated 2.4.1
- Section 3.2: added 3.2.1: submission of hourly data is mandatory, updated 3.2.5 on data latency
- Section 3.3: updated 3.3.1
- Section 3.6: updated 3.6.1, updated 3.6.6 for RINEX header
- Section 3.7: updated 3.7.4, added 3.7.6

May 15, 2007: Section 3.7.4: update of mandatory messages

March 1, 2007: Section 3.7: update of mandatory messages

Dec. 5, 2006:

- Added guidelines for stations streaming real-time data
- Promote usage of multi-GNSS equipment
- Recommendation for new antennae/radomes or antenna/radome replacements to be absolutely calibrated

*Very strict rules are inconsistent with the voluntary nature of the EUREF Permanent Network (EPN). However, stations participating to the EPN must agree to adhere to certain standards and conventions, which ensure the quality of the EPN. This document lists the **requirements** that all **EPN Tracking Stations (TS)** must follow, as well as some additional desirable characteristics, which are not mandatory, but enhance a station's value to the EPN.*

For historical reasons, existing EPN stations may not adhere to all guidelines described in this document. However, it is expected from station managers and operating agencies that when upgrading equipment at an existing EPN station, this new equipment will be conform to the guidelines.

For proposed EPN stations please follow the stepwise "Procedure for Becoming an EPN station": https://epncb.oma.be/documentation/guidelines/procedure_becoming_station.pdf (which includes a reference to this document).

For IGS tracking stations, please refer to "IGS Site Guidelines" <https://igs.org/station-resources/#site-guidelines> available from the IGS Central Bureau.

In the following text, we distinguish site and station as follows:

- A site is a local geographical area where one or more geodetic markers are available. Each of the geodetic markers can be characterised by an IERS DOMES number (required for an EPN station). The first five numbers of the DOMES number of markers located on the same site are identical.*
- A station observes at a unique geodetic marker (and DOMES number) at a site with a specific receiver and antenna. Multiple stations can exist at one site, and even on one geodetic marker (e.g. two receivers connected to the same antenna).*
- Using RINEX 2 conventions, a station is characterized by a unique 4-character identification (ID) and is occupied with a specific GNSS receiver and antenna. In cases where multiple GNSS receivers are split off from the same antenna, each GNSS receiver is considered a separate station and a different 4-character ID needs to be defined. The first four characters of the RINEX data files correspond to the station 4-character ID. Each station is documented using a unique station log. The 'Surveyed Local ties' section in the station site log describes how different geodetic markers at the station's site are related to each other.*

CONTENTS

GUIDELINES FOR EPN STATIONS	1
1 ORGANIZATION OF THE EPN DATA FLOW	5
1.1 NETWORK COMPONENTS.....	5
2 STATIONS SETUP	6
2.1 EQUIPMENT	6
2.2 MARKER AND MONUMENT	8
2.3 COLLOCATED INSTRUMENTS, MARKERS AND LOCAL TIES.....	8
2.4 LINK WITH THE UELN AND ETRS89	9
2.5 NEW STATIONS.....	10
2.6 STATION LABELS.....	10
3 STATION OPERATION	12
3.1 RESPONSIBILITIES.....	12
3.2 FORMAT AND DISTRIBUTION OF HOURLY AND DAILY DATA.....	13
3.3 FORMAT AND DISTRIBUTION OF REAL-TIME DATA	16
3.4 FORMAT AND DISTRIBUTION OF HIGH-RATE DATA.....	18

1 ORGANIZATION OF THE EPN DATA FLOW

1.1 Network Components

En route to Analysis Centres and other users, the tracking data collected by permanent GNSS receivers flow through the following components of the EPN network:

- **Tracking Station (TS):** Permanent GNSS tracking receivers and antennae on suitable geodetic markers.
- **Station Owner (SO):** Listed as the “Responsible Agency” in the station site log, the Station Owner holds overall responsibility for the operation of the station and its integration into the EPN. This includes: validating the GNSS data, converting raw data to the Receiver Independent Exchange Format (RINEX), compressing data, generating real-time streams, uploading data to the EPN Data Centres and Broadcasters via the Internet, and maintaining station metadata, such as the site log, data license, and other relevant documentation.
- **M³G:** The Metadata Management and Distribution System for Multiple GNSS Networks (M³G) is the official platform used to manage and distribute metadata for EPN stations. It allows authorized agencies to enter, update, and publish station metadata, such as site logs, data licenses, and DOI-related information.
- **Metadata Maintainer (MM):** The agency with authorized access to M³G and which has the operational responsibility for maintaining the station's metadata in the system. In most cases, the MM is the same agency as the Station Owner. However, operational maintenance of the metadata may be delegated to an external agency. Regardless of delegation, final responsibility remains with the Station Owner.
- **Local Data Centre (LDC):** Collects the hourly and daily RINEX data of all stations in a local network and distributes them to the users (local and EPN). The LDC will forward the data (or a selection) of the local stations to the Regional Data Centres. If there is no LDC available for a particular station, then its data will flow directly to the Regional Data Centres.
- **Regional Data Centre (RDC):** It collects the hourly and daily RINEX data from all EPN stations. The RDC makes the data available to the local, regional and IGS users. To guarantee the availability of the EPN data at the regional level, two RDCs are in operation, see https://epncb.oma.be/networkdata/data_access/dailyandhourly/datacentres.php.
- **Local Broadcaster (LB):** It receives the real-time data streams from the stations in a local network and disseminates them, without changing them, on request to clients. Clients may be users, monitoring tools, Regional Broadcasters, Data Centres, or Analysis Centres.
- **Regional Broadcaster (RB):** It receives all the EPN real-time data streams and disseminates them, without changing them, on request to clients. Clients may be users, monitoring tools, other broadcasters, Data Centres, or Analysis Centres.
- **High-Rate Data Centre (HRDC):** It collects 1 Hz data from all EPN stations (either directly from the stations (recommended) or from real-time streams), archives them in 15-min files in RINEX format and makes them available to the users.
- **Historical Data Centre (HDC):** It provides an open access historical archive of all daily RINEX EPN data, including data from EPN stations prior to their installation in the EPN. RINEX headers are corrected to be consistent with the station metadata (station site log).

In the following, we will use the wording “Data Centre” (DC) for both the RDC and LDC, and “Broadcaster” for both the LB and RB.

2 STATIONS SETUP

EPN tracking stations should fulfil the guidelines below.

2.1 Equipment

Receiver

The GNSS receiver must be set to

- 2.1.1 Track satellites down to a cut off of 0°.
- 2.1.2 Track as many satellites, healthy and unhealthy (all-in-view tracking), from as many constellations as possible (within receiver limitations), always including all GPS satellites as a minimum.
- 2.1.3 Record at least dual frequency phase and code measurements from all GNSS.
- 2.1.4 Disable pseudorange and/or phase smoothing. If activated by mistake, any changes in the receiver setting shall be reported in the station site log.
- 2.1.5 Synchronize the actual instant of observation with true GPS time within ± 1 millisecond of the full second epoch.

Additionally desired characteristics are:

- 2.1.6 The receiver must be capable of providing GLONASS and Galileo measurements (or easily upgradeable via firmware) in addition to GPS.
- 2.1.7 The receiver must be capable of providing third frequency measurements in the RINEX data files for all constellations.
- 2.1.8 The receiver must be set to provide signal-to-noise values for all tracked signals in the RINEX files at the highest precision possible.
- 2.1.9 The logging of Doppler observations is recommended.
- 2.1.10 The receiver and ideally other station equipment, such as computers, should be protected against power failures by providing surge protection and backup power wherever feasible.
- 2.1.11 The receivers shall be upgraded with firmware upgrades from the manufacturer within 6 months of the firmware publication. Firmware updates shall be noted in the station site log, and announced via a EUREF mail.

Antenna and Radome

- 2.1.12 Avoid using radomes unless required operationally, for instance due to weather conditions, antenna security, wildlife concerns, etc.
- 2.1.13 The antenna (+radome, if any) must be known to the IGS and EPN; a standard name must be designated for the antenna and radome in https://epncb.oma.be/pub/station/general/rcvr_ant.tab.

- 2.1.14 The antenna (+radome, if any) must be included in the EPN phase centre variation file (see https://epncb.oma.be/documentation/equipment_calibration/) with zenith and azimuth-dependent absolute calibration values down to the horizon.
- 2.1.15 The antenna must be levelled and oriented to True North using the defined antenna reference marker (see vendor instructions, contact the Central Bureau with questions).
- 2.1.16 The antenna must be rigidly and securely attached to the top of the station monument.
- 2.1.17 The eccentricities (easting, northing, height) from the marker to the antenna reference point (ARP, defined for each antenna in <https://epncb.oma.be/pub/station/general/antenna.gra>) must be surveyed and reported in the station site log and RINEX headers to ≤ 1 mm accuracy.
- 2.1.18 The antenna height eccentricity with respect to the marker must be less than 10 m, the horizontal eccentricities must be below 5 m (see also 2.1.21).
- 2.1.19 The antenna radome, if used, should be spherical, mounted concentrically over the average phase centre height of the antenna and properly aligned following the manufacturer instructions.

Additionally desired characteristics are:

- 2.1.20 The antenna must be capable of tracking new GPS, GLONASS, and Galileo signals (even if the receiver is not (yet) capable to do so).
- 2.1.21 Horizontal eccentricities (northing and easting) of 0 are preferred. The results of repeated local ties, connecting the antenna reference point to the marker must be carefully documented.
- 2.1.22 It is recommended to clean the antenna (without changing its position or removing it!) at least once a year. During wintertime, remove snow coverage as frequently as possible. Report the cleaning in the station site log.
- 2.1.23 If a radome must be used, radomes uniformly manufactured are preferred.
- 2.1.24 Absolute individual antenna calibrations of an antenna are preferred. New individual absolute antenna calibrations for an antenna+radome pair have to be sent to EPN CB prior to installing the antenna+radome pair.
- 2.1.25 If an uncalibrated antenna/radome (see <https://epncb.oma.be/pub/station/general/calibration.txt>) is removed, please make it available to a calibration facility for individual calibration and send the calibration to the EPN CB.

Operation

- 2.1.26 Stations must be planned and installed for continuous and permanent operation.
- 2.1.27 Site location must be secured over the long-term so that changes of antenna location at the site are highly unlikely in the foreseeable future (no planned construction, demolition, etc in the site vicinity).
- 2.1.28 Minimally, the station must provide 24hr daily files at 30-sec sampling, delivered as soon as possible after 23:59:30 for the previous day

Additionally desired characteristics are:

- 2.1.29 In case a major equipment upgrade is foreseen at a well-performing station, e.g. upgrading from GPS-only to multi-GNSS tracking capability, then install the new equipment on a new monument

and propose the station to the EPN. Keep the old station simultaneously in the EPN as long as possible.

- Ample, reliable power and communications (preferably Internet) to enable reliable data transfer
- Physical site security, appropriate to local necessity.

2.2 Marker and Monument

- 2.2.1 The station permanent marker shall be appropriate to allow the assignment of an M-type IERS DOMES number, to indicate the antenna is referenced to a physical point on a monument or pillar (see: <https://itrf.ign.fr/en/network/domes/request>).
- 2.2.2 The marker should fulfil standard requirements for a first order geodetic monument with respect to stability, durability, long-term maintenance, documentation and access. The marker description should be fully documented in the EPN station site log file.
- 2.2.3 Obstruction should be minimal above 5 degrees elevation, but satellite visibility at lower elevations is encouraged whenever possible (see 2.1.1).
- 2.2.4 Signal reception quality has to be verified, especially with respect to interference of external signal sources like radars, and with respect to multipath.
- 2.2.5 The antenna must be setup to minimize code and phase reflections (multipath), by mounting it away from close reflecting surfaces or by applying some passive protection directly below the antenna (microwave absorbing material, etc.).
- 2.2.6 Local ties to other markers on the sites should be determined in the ITRF coordinate system to guarantee 1-mm precision in all three dimensions. Offsets are given in delta-X, delta-Y, delta-Z, where X, Y, Z are the geocentric Cartesian coordinates (ITRF).
- 2.2.7 The station should not be moved to a different monument unless absolutely necessary. Moving to a new monument would require that a new station be established with a separate station ID and log.
- 2.2.8 If upgrading a station from a bad monument design to a better antenna mount, involving moving the original marker, then the upgrade is essentially the commissioning of a new station, so all guidelines are fully applicable (see '2.5 New Stations').
- 2.2.9 When moving a station to a different monument at the site is unavoidable, parallel operations are required for as long as possible (3 months or more if possible) between the old and new station positions.

2.3 Collocated Instruments, Markers and Local Ties

- 2.3.1 Additional monuments (preferably at least three) are desirable for surveys, testing and operating equipment in parallel during upgrades, but it is preferable to maintain one station as the principle station for the EPN.
- 2.3.2 Co-location with other geodetic techniques such as SLR, VLBI, DORIS or gravimeters is highly desired.
- 2.3.3 Precise meteorological instrumentation is encouraged:
- Data are to be prepared in RINEX "m" files. See the RINEX specifications, <https://epncb.oma.be/documentation/formats/rinex.php>.

- RINEX 'm' data is to be transmitted on the same schedule as the RINEX observation files (hourly for hourly stations; otherwise daily).
- Data sampling interval must be less than 60 minutes and is recommended to be 10 min.
- Pressure sensor accuracy must be within 0.5 hPa.
- Temperature sensor accuracy must be at least 1 Kelvin.
- Drift and bias must be minimized.
- Temperature effects on the pressure measurements should be minimized, e.g. with solar shielding or by placing the sensor in a nearby building if necessary.
- Measurement of the instrument height in relation to the GNSS antenna mark must have an accuracy of 1 m or better.
- Meteorological instruments are to be calibrated periodically according to the manufacturer's recommendations to assure the above accuracy specifications are met. The calibration dates should be included in the station site log.

2.3.4 Other scientific systems which rely on accurate positioning, such as timing labs, are also recommended where appropriate.

2.3.5 The 3-dimensional local ties between the GNSS marker, co-located instrumentation (e.g. DORIS, SLR, VLBI, gravity, tide gauge, levelling) and other monuments should be re-surveyed as frequently as practical (ideally each 2 years) to an accuracy of 1-mm and reported in ITRF:

- The marker → antenna reference point eccentricities should be re-verified during such a survey.
- Repeat the survey after known motion incidents such as earthquakes.
- All survey data, but especially ties to other IERS/IGS/EPN markers, should be rigorously reduced in a geocentric frame related to ITRF (preferably ITRF itself) and the results be made available in SINEX format (defined at), including full variance-covariance information.
- Survey notes and intermediate results of the 3D ties shall be preserved and made available publicly. Local tie survey results involving other space geodetic technique instruments should be sent to the IERS/ITRS product centre (<https://itrf.ign.fr/en/local-ties/survey-metadata>).
- For the local ties, the guidelines derived by the European Combined Geodetic Network (ECGN) "ECGN Standards for Local Ties" (https://epncb.oma.be/documentation/guidelines/ECGN_local_ties_standards.pdf) should be followed.

2.4 Link with the UELN and ETRS89

It is recommended to

- 2.4.1 Connect the GNSS marker to the UELN and the national precision levelling network either using precise levelling or a local GNSS campaign. For the levelling connection, the ECGN guidelines "ECGN Standards for the Levelling Connection of the ECGN Station" (https://epncb.oma.be/documentation/guidelines/ECGN_levelling_standards.pdf) should be followed.
- 2.4.2 Integrate the station marker in the national official densification of the ETRS89. For performing national densifications of the ETRS89, the "Guidelines for EUREF densifications" (https://epncb.oma.be/documentation/guidelines/Guidelines_for_EUREF_Densifications.pdf) should be followed.

2.5 New Stations

In addition to sections 2.1 - 2.4, new stations should respect the requirements below. The EUREF Governing Board (GB) will evaluate the benefits of a proposed EPN station.

- 2.5.1 New stations must provide an added-value to the EPN, according to the objectives of EUREF.
- 2.5.2 New stations can only be integrated in the EPN after following the 'Procedure for Becoming an EPN Station', see https://epncb.oma.be/documentation/guidelines/procedure_becoming_station.pdf
- 2.5.3 New stations must have equipment capable of
 - tracking all GPS, GLONASS and Galileo signals (or easily upgradeable via firmware)
 - providing third frequency measurements in the RINEX data files (L5) for all constellations
 - providing real-time data using the Ntrip technology, preferably in RTCM 3.x or, in particular RTCM HP-MSM (see section 3.3 for explanation).
- 2.5.4 Set to provide signal-to-noise values for all tracked signals: S1, S2, etc in the RINEX files at the highest precision possible.
- 2.5.5 Agencies are encouraged to select potential new stations whose station monument
 - is conform with current best practices observed by principal geodetic agencies. A drilled-braced tripod structure or tapered pillar type monument are typically constructed. Roof or structure mounted antennas should be avoided.
 - foundation shall extend to bedrock or be deeply embedded into the stable subsurface and isolated from surface effects where bedrock is not accessible.
 - is on a stable regional crustal block, away from active faults or other sources of deformation, subsidence, etc.
 - is on firm, stable material, preferably basement outcrop.
 - is not located on soil that might slump, slide, heave, and does not vary in elevation (e.g. because of subsurface liquid variations).
- 2.5.6 The station location should not
 - suffer from significant changes to the surroundings (changes to buildings or trees; new construction, etc...) foreseen or likely
 - have excessive radio frequency interference
 - have excessive RF reflective surfaces (fences, walls, etc.) and other sources of signal multipath
 - have excessive natural or man-made surface vibrations from ocean waves or heavy vehicular traffic.
- 2.5.7 Co-location with other geodetic techniques such as SLR, VLBI, DORIS, absolute or superconducting gravimeters, Earth tide gravimeters, seismometers, strain meters, InSAR reflectors and ocean tide gauges are also desirable and will enhance the value of the station for multi-disciplinary studies.

2.6 Station Labels

- 2.6.1 Station operational status:

Based on the information provided by station managers in M³G, an EPN station can have one of the following operational statuses:

- **Active:** The station site log indicates that both a receiver and antenna are currently installed, and the station is expected to deliver data to the EPN. This label is assigned automatically by M³G.
- **Inactive:** The station site log shows that either the receiver or antenna is not currently installed. This status is also assigned automatically by M³G.
- **Decommissioned:** An inactive station can be marked as decommissioned by the station manager in M³G if the station site log confirms that both the receiver and antenna have been removed and the station will not be reactivated (for example because the monument has been destroyed).

The up-to-date list of active, inactive, or decommissioned stations can be consulted online from <https://gnss-metadata.eu/site/stations> (column "STATUS")

2.6.2 Data availability status:

Dormant: Based on data availability, a station is labelled dormant if it has not provided data to the EPN for more than three months. This label is assigned automatically by the EPN Central Bureau. The list of dormant EPN stations is updated daily and is available from https://epncb.oma.be/pub/station/general/dormant_stations.txt

2.6.3 Integration status within the EPN:

Stations are also classified based on their integration status within the EPN:

- **Proposed:** A station that has been submitted for inclusion in the EPN but does not yet meet all requirements. The list of proposed EPN stations is updated daily and is available from <https://epncb.oma.be/networkdata/proposed.php>.
- **Included:** A proposed station that has fulfilled all criteria and has been formally integrated into the EPN. The list of stations included in the EPN is updated daily and is available from <https://www.epncb.oma.be/networkdata/stationlist.php>.
- **Former:** A station that has been decommissioned or has been removed from the EPN by decision of EUREF due to non-compliance or other reasons. Data from former stations are no longer monitored by the EPN Central Bureau. The list of former EPN stations is updated daily and is available from https://epncb.oma.be/pub/station/general/former_stations.txt

The statuses Proposed, Included, and Former are assigned manually by the EPN Central Bureau. The procedure for proposing a station and the criteria for its inclusion are described in [Procedure for becoming an EPN station](#).

2.6.4 Usage in the combined EPN solution:

Upon generating the daily combined EPN solution, the EPN Analysis Center Coordinator (ACC) may detect stations whose coordinates deviate significantly from the expected values. The ACC maintains a list of these stations, specifying the periods they should be excluded from the EPN analysis. The list is available from https://epncb.oma.be/pub/station/general/sites_excluded.txt.

3 Station Operation

The agency owning the GNSS station is responsible for properly maintaining the station and ensuring it is operated according to EPN guidelines. In some cases, this station owner may delegate part or all of the operational tasks—such as metadata maintenance in M³G, data archiving, or data distribution—to an external partner. Although certain responsibilities may be delegated, the station owner retains overall responsibility for the station.

3.1 Responsibilities

Data Archiving, Quality and Distribution

The agency ensuring the station operation must

- 3.1.1 Ensure that the GNSS equipment, and its surroundings, is not disturbed or changed unless a clear benefit outweighs the potential for discontinuities in the time series. Examples include: equipment failure, planned upgrade of obsolete equipment or vendor-recommended firmware updates
- 3.1.2 Archive the station's data (preferably in its native format) in case needed for data recovery or engineering purposes
- 3.1.3 Download raw data from the receivers of the local network and reformat them into the agreed-upon exchange format (according to sections 3.2 and 3.3)
- 3.1.4 Check on regular basis the results of GNSS data quality checks (e.g. performed by the EPN CB) to detect potential performance degradations of your station.
- 3.1.5 Take appropriate action if the station performance degrades, e.g. the alert/engagement of on-site staff
- 3.1.6 Ensure full responsibility for reliable data handling and transmission to the relevant Data Centres and real-time Broadcasters, if any (according to sections 3.2 and 3.3).
- 3.1.7 Correct the meta-data in the RINEX header (see 3.2.7), real-time streams (see 0) or station site log (see 3.1.10) as soon as possible, if an inconsistency advisory is received from the CB.
- 3.1.8 Must maintain full capability to repair, upgrade and maintain the station and its software systems, even if the original engineers are no longer available.

Station Site Logs

The station owner must ensure an agency is accepting responsibility to

- 3.1.9 Document changes at the station (or its environment) by submitting an UPDATE station site log, within one business day, to the Central Bureau, using the "Metadata Management and distribution system for Multiple GNSS Networks (M³G)" available from <https://gnss-metadata.eu/>.
- 3.1.10 Correct the station site log as soon as possible, if an advisory of station site log inconsistencies is received from the CB.

Station Pictures

The agency accepting responsibility for proper station operation must

- 3.1.11 Provide pictures of the antenna installation in the 4 cardinal directions (as a minimum), the monument and its vicinity; update them after configuration changes on M³G (<https://gnss-metadata.eu/>).

Individual Antenna Calibrations

If available, the agency accepting responsibility for proper station operation can optionally

- 3.1.12 Submit individual antenna calibration files to the EPN CB to be stored in an archive containing individual antenna calibrations of EPN stations.

EUREF mail

The EUREF mail exploder notifies the EUREF community through E-mail about events related to the EUREF Permanent Network, for example: station configuration changes, abnormal stations conditions, analysis results, etc.

Subscription information is available from

https://epncb.oma.be/newseventslinks/mail_rulings.php?community=EUREF

The agency accepting responsibility for proper station operation must

- 3.1.13 Subscribe to EUREF mail
- 3.1.14 Follow EUREF mail on a regular basis to remain informed on EPN developments.
- 3.1.15 Send (in advance if possible, otherwise within one business day) an advisory EUREF mail in the following cases:
- Station status messages which can affect position solutions or which requires a station site log update:
Changes in antenna, radome, receiver, cabling, frequency standard, receiver settings (such as elevation cut off angle), or environment (such as tree removal or building construction).
 - If a RINEX file must be resubmitted to a DC due to corruption, incorrect metadata, etc.
 - If a station is expected to be unavailable for more than one week.
 - When a station is decommissioned permanently.
 - When a problem or error in the station or its station site log is discovered and corrected. Briefly describe in the message what was changed.
 - Avoid using personal information in the emails and preferably sign the emails as e.g., *GNSS team of Agency X*
 - Place the 9-character station ID (as used in RINEX 3) in the Subject line.

3.2 Format and Distribution of Hourly and Daily Data

EPN stations make their hourly and daily RINEX data routinely available at two DCs. The names of these DCs are indicated in the station site log as the “Primary Data Centre” and “Secondary Data Centre”. Based on this principle, each station/OC can distribute its data following one of the three standard data flow schemes given in Figure 1.

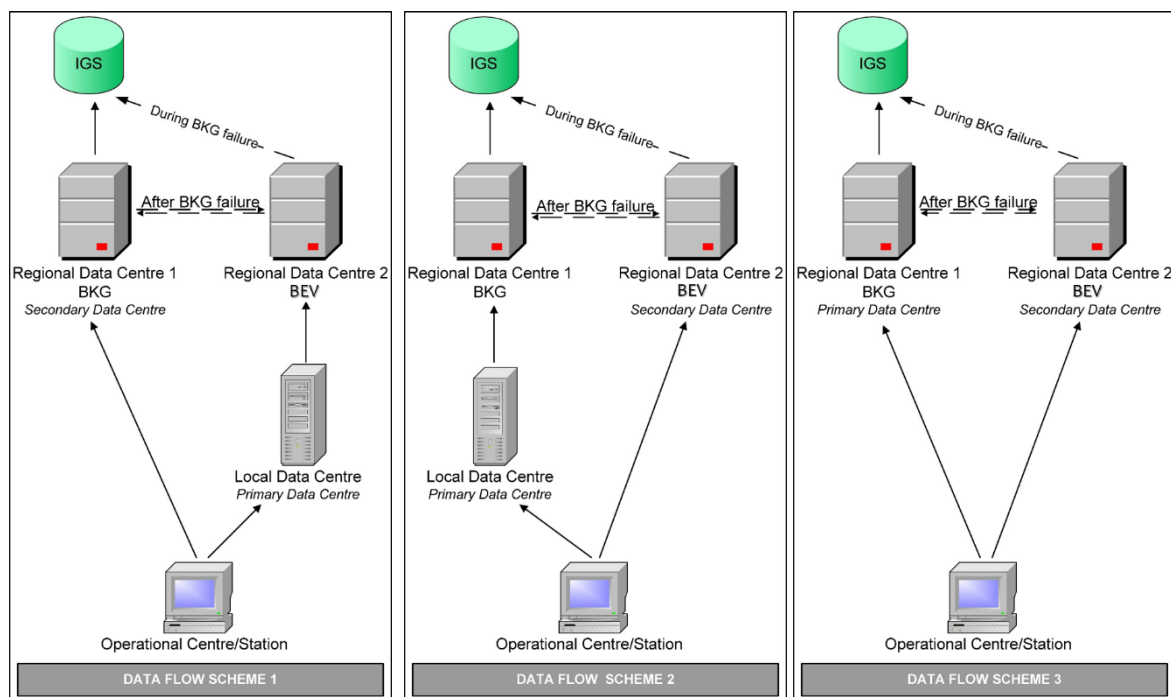


Figure 1: Standard data flow schemes for hourly and daily data within the EPN.

The majority of the EPN stations/OCs use scheme 3 and directly provide their data to the two RDC. If however a station chooses to use scheme 1 or 2, then it has to make sure that the LDC to which it is sending its data are effectively uploading the data routinely to the agreed RDC. Exceptionally, if none of the schemes in Figure 1 can be applied, the scheme given in Figure 2 can be used. In this scheme, the station/OC submits its data only to one LDC, noted as the 'Primary Data Centre' in the station site log. In case of a failure of this LDC, the station data flow to EUREF will be interrupted. For this reason, this last scheme is not considered as optimal.

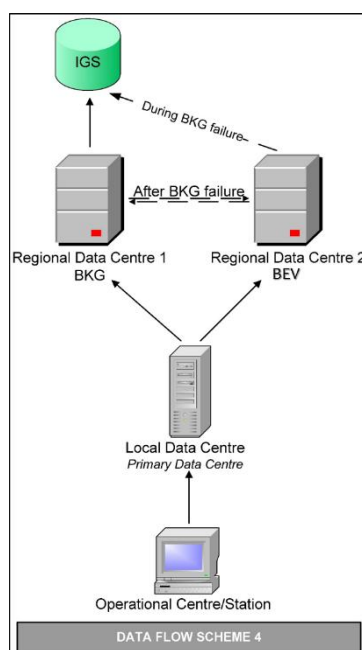


Figure 2: EPN data flow scheme for hourly and daily data, used only in case the standard data flow schemes cannot be implemented.

- 3.2.1 The station must provide hourly and daily observation files to the relevant Primary and Secondary Data Centres (as indicated in the station site log) in the RINEX format. Check the currently used standard RINEX format from <https://epncb.oma.be/documentation/formats/rinex.php>.
- 3.2.2 RINEX data files must be directly generated from the receiver's native data files (not from real-time data streams), and tagged in GPS time.
- 3.2.3 RINEX 3 or 4 (called 3+) are the mandatory data exchange formats. RINEX 2 data are only allowed in for historical data. It is not allowed to convert a RINEX 2 file into RINEX 3+. Repeated switching between RINEX 2-only or RINEX 3+-only submissions is not allowed.
- RINEX Observation files will normally be exchanged in the Hatanaka Compact form.
 - All observation files are then gzipped (.gz) for transmission to the DCs.
 - The RINEX navigation and meteo files are prepared in gzipped form.
 - File naming conventions set forth in the RINEX 2 and 3+ format descriptions, Section 4 "The Exchange of RINEX files" will be followed.
- 3.2.4 Daily and hourly observation data must be distributed at 30-second intervals aligned to :00 and :30.
- 3.2.5 The daily observation files contain the observations collected between GPS time 00:00:00 and 23:59:59.
- 3.2.6 The hourly observation files data contain the observations collected between GPS time 00:00 and 59:59 of each hour.
- 3.2.7 RINEX observation header information, especially the 9-character station ID, receiver and antenna information, DOMES number, and antenna eccentricities, must be up-to-date and must match the information in the station site log. The following conventions are used:
- The 9-character station ID must be found in the "MARKER NAME" field.
 - The DOMES number must appear in the "MARKER NUMBER" field.
 - The receiver serial number, type and firmware must be found in the "REC # / TYPE / VERS" fields.
 - The antenna serial number, antenna type and radome type (if no radome is used, use "NONE") must be found in the "ANT # / TYPE" fields.
 - The RINEX headers must begin showing an equipment change as near as possible to the actual time of the change.
 - Use only generic team names or emails (and replace "@" by "at") in the RINEX header, for example:

GNSSatROB	Royal Observatory of Belgium	OBSERVER / AGENCY
-----------	------------------------------	-------------------
 - For RINEX 4 headers, it is strongly recommended to include the Digital Object Identifier (DOI) and data license associated with the RINEX data, as well as a Persistent Identifier (PID) linking to the station metadata. For example,

https://doi.org/10.24414/ROB-GNSS-BRUX00BEL CC-BY-4.0 https://gnss-metadata.eu/MOID/BRUX00BEL	DOI LICENSE OF USE STATION INFORMATION
---	--
- 3.2.8 The daily RINEX navigation message file contains all messages with TOC/TOE (time of clock, time of ephemeris) between 00:00 and 23:59 GPS time of the respective day.
- 3.2.9 The station's data handling and transmission to the relevant Primary and Secondary Data Centres by the station operator must occur reliably as scheduled (within 10 minutes after the last recorded epoch for daily files and 5 minutes for hourly files) using automated procedures and verified to be uncorrupted and complete in order to avoid resubmissions.

- 3.2.10 After a communication outage between the station and the OC, or between the OC and the DCs, all recovered data files must be submitted to the DCs as quickly as possible. At least a retry should be done within the hour. The usage of automated procedures is strongly recommended. Missed hourly data files should be transmitted only for files less than 3 days old. A EUREF mail shall be sent explaining the outage and the data availability over the period.
- 3.2.11 In case of reliable hourly data upload, it is possible to discontinue the submission of the daily files. The following procedure must be adopted:
- 1) Send an e-mail to the DC (with EPN CB in cc) some days before you intend to cancel the transfer of daily files announcing that intention and ask the DC to concatenating the hourly files to daily ones
 - 2) Wait for the confirmation of the DC
 - 3) If the DC agrees, stop the transfer of daily files

Additionally desired characteristics are:

- 3.2.12 The submission of mixed navigation files (MN extension for RINEX 3) is recommended instead of submitting a separate navigation file for each constellation.
- 3.2.13 The signal-to-noise observables (S1, S2, etc...) should be included in RINEX files (see 2.1.10).
- 3.2.14 Stations equipped with high-precision weather sensors are encouraged to submit the daily RINEX meteo files (see 2.3.3).

3.3 Format and Distribution of Real-time Data

In addition to the standard (hourly and daily) data flow, stations are encouraged to make their data available in real-time using the Ntrip-technology. The current version Ntrip 2.0 is described in RTCM Paper 111-2009-SC104-STD, see <http://www.rtcn.org> - with charge, <https://igs.bkg.bund.de/ntrip/about>). Similar to the RINEX hourly and daily files, these real-time streams are generated from the receiver's native data.

Stations streaming real-time data make their data available at the Regional Broadcasters who make available all EPN real-time data streams. If the station is part of a local network, it can also make available its real-time data at a Local Broadcaster. This data flow scheme, therefore, corresponds to the scheme given in Figure 2.

- 3.3.1 Real-time data should be sent to (at least) two independent Regional Broadcasters listed in https://epncb.oma.be/networkdata/data_access/real_time/broadcasters.php, one of them being the Regional Broadcaster at BKG. The real-time data transfer should be done using the receiver's software or, if not available, the Ntrip-server software.
- 3.3.2 Alternatively, station operators can make the data available so that it can be pulled by the Regional Broadcasters, either from a Local Broadcaster or from the receiver itself.
- 3.3.3 Real-time station operators must strive to meet the following objectives:
- The station Internet service should be sufficient to meet the minimum performance standards required to send 1-Hz data to two or more real-time broadcasters. A latency of two seconds or less from station to data centre is acceptable.
 - Stream data should not affect completeness of the file-based data sets. Hourly and daily files from streaming stations should contain all the data from each epoch and all epochs within the file period.
- 3.3.4 The station operator needs to make the station's most complete and relevant RTCM stream available in the following format: SSSSMRCCC0, e.g., BRUX00BEL0 (see for example

https://files.igs.org/pub/data/format/rinex_4.02.pdf, Table A1 for description). Please note, that a station operator can provide more than one stream to the caster. It is expected that the most complete stream will be addressed with the 10th character as "0".

- 3.3.5 The format of the data stream should be RTCM 3.x. It is recommended to use RTCM MSM7 (Multiple Signal Messages) as these messages provide the most precise observations. If the transmission bandwidth is limited, the usage of MSM5 is acceptable. Information on RTCM-MSM can be found at <http://www.rtcn.org> (with charge).
- 3.3.6 After a communication outage or reception of a NABU message, the data flow should be restored as quickly as possible, preferable using an automated procedure.
- 3.3.7 It is recommended that message types containing metadata (1005, 1006, 1007, 1008, 1033, see Table 2) are streamed with an update rate of at least 60 seconds or higher. Provide at least message 1006 (use 1005 only if 1006 is not available) and 1008 (use 1007 only if 1008 is not available) or message 1033.

Message type	Content
1005	(X, Y, Z) coordinates of the antenna reference point
1006	(X, Y, Z) coordinates of the antenna reference point + height of antenna reference point above marker
1007	Antenna and radome type description
1008	Antenna and radome type description + antenna serial number
1033	Receiver and antenna type and antenna serial numbers

Table 2: RTCM 3.x message types providing station metadata

- 3.3.8 It is expected that the observation messages (see Table 3 and 4) are updated with a frequency of 1 Hz.

Message type	Content
1003	GPS code and carrier phase observations
1004	GPS code and carrier phase observations + CNR (carrier-to-noise ratio) + code ambiguity
1011	GLONASS code and carrier phase observations
1012	GLONASS code and carrier phase observations + CNR (carrier-to-noise ratio) + code ambiguity
1019	GPS satellite broadcast ephemeris
1020	GLONASS satellites broadcast ephemeris

Table 3: RTCM 3.x (x≤1) message observation types

Message type	Content
1071	Compact GPS Pseudo ranges
1072	Compact GPS Phase ranges
1073	Compact GPS Pseudo ranges and Phase ranges
1074	Full GPS Pseudo ranges and Phase ranges plus CNR
1075	Full GPS Pseudo ranges, Phase ranges, Phase range rate and CNR
1076	Full GPS Pseudo ranges and Phase ranges plus CNR (high resolution)
1077	Full GPS Pseudo ranges, Phase ranges, Phase range rate and CNR (high resolution)

Table 4: RTCM 3.x (x≥2) MSM message types for GPS

- 3.3.9 Stations still streaming RTCM 3.0 or 3.1 should stream observation messages 1004 (above 1003) and 1019 for GPS and 1012 (above 1011) and 1020 for GLONASS.
- 3.3.10 Stations streaming RTCM 3.2 ($x \geq 2$) should stream additional information (all messages since RTCM 3.0 plus applicable observation and ephemeris data of the tracked satellite systems. Table 4 provides example MSM message types for GPS; the corresponding message ranges for GLONASS (1081–1087), Galileo (1091–1097), SBAS (1101–1107), QZSS (1111–1117), and BeiDou (1121–1127) follow the same structure.
- 3.3.11 RTCM code and phase observations shall not be corrected to refer to the antenna reference point (this correction is typically known as the antenna phase centre correction). If the correction is applied, the antenna type in the stream is typically set to NULLANTENNA (message types 1007/1008 for RTCM 3.x).
- 3.3.12 The metadata reported in the stream (receiver, antenna/radome, antenna height) must agree with the information provided in the EPN station site log. It is encouraged that the most recent ETRS89 coordinates, in line with the latest national realisation of the ETRS89 or the ETRS89 coordinates maintained by the EPN Reference Frame Coordinator <https://epncb.oma.be/pub/station/coord/EPN/RTCM.CRD>). Take care on the coordinates provided within the stream (message types 1005/1006): they should be the coordinates of the Antenna Reference Point (and not the Marker)!

3.4 Format and Distribution of High-Rate Data

Some applications may profit from 1 Hz GNSS data with low latency.

- 3.4.1 Stations are encouraged to upload 15-min RINEX 3 data with 1 Hz sampling rate to the HRDC. The files should contain the data for mm:ss 00:00-14:59, 15:00-29:59, 30:00-44:59, 45:00-59:59, resp., and they should be sent to the HRDC directly after the closing of the RINEX file to ensure the low latency.
- 3.4.2 These files can be generated as follows (in order of preference):
- 1) directly from the receiver by converting the receiver's binary data to RINEX files. Following the RINEX 3 file naming convention, the data source is "R".
 - 2) by converting the station's real-time data stream in the receiver's native format to RINEX files. Following the RINEX 3 file naming convention, the data source must be set to "S".
 - 3) by converting the station's real-time data stream, available in RTCM MSM format, to RINEX files. Following the RINEX 3 file naming convention, the data source must be set to "S".
- 3.4.3 If the station is not capable of generating high-rate RINEX 3 files, then the HRDC, that is also operating a real-time broadcaster, will convert the station real-time stream to 1-Hz RINEX 3 files with each 15 minutes of data. Following the RINEX 3 file naming convention, the data source must be set to "S". More info in "Guidelines for Data Centres and Broadcasters".

Acknowledgments

This document was assembled from many preceding documents, including the previous version of "Standards for EPN Stations and Operational Centres", "Procedures for Becoming an EPN Station" and the latest version of "IGS Site Guidelines". Thanks are due to EUREF colleagues that have provided suggestions on this version.



This document is maintained by the EPN Network Coordinator at the EPN Central Bureau. Please direct comments, questions, and suggestions to epncb@oma.be.